

InRisk Labs | Quantitative Analyst Case Study

Submitted By: MUNI KUMAR CHEMURU

Candidate Email: munikumarchemuru2003@gmail.com

Role Target: Quantitative Analyst

Date of Submission: September 2026

GitHub Repository: https://github.com/munikumar2003/inrisk_labs_qa

Project Deployment: <https://inrisklabsqa.vercel.app/>

EXECUTIVE SUMMARY

This report designs, prices, and evaluates a commercial parametric weather insurance contract protecting homeowners and retail financiers against solar generation deficits on residential 3.0 kVA rooftop solar installations in Ahmedabad (Pincode 380006).

Using 20 years of ECMWF **ERA5-Land reanalysis climate data (2005–2024)**, we establish a physics-grounded, temperature-derated Global Horizontal Irradiance (GHI) index. Setting the **trigger threshold at 90% of AEP50 (4,140 kWh)** and an **exit floor at 70% of AEP50 (3,220 kWh)** produces:

- **Pure Burn Cost:** ₹75 / year (empirical expected annual loss)
- **Actuarial Loading:** 55% (25% Climate Volatility, 15% NBFC/Admin, 15% Reinsurer Margin)
- **Commercial Premium:** ₹116 / year (₹137 with 18% GST)
- **Monthly Loan EMI Embedded Cost:** ₹11 / month (+1.8% on typical ₹3,200 loan)
- **Loan Solvency Coverage:** Guarantees 1.0x Debt Service Coverage Ratio (DSCR) during catastrophic monsoon years.

TASK 1: WEATHER INDEX SELECTION & SCIENTIFIC JUSTIFICATION

1. Photovoltaic Conversion Physics

Solar energy generation for grid-connected PV is governed by the standard single-diode conversion relationship:

$$E_d = P_{\text{STC}} \times \left(\frac{GHI_d}{G_{\text{STC}}} \right) \times PR \times [1 - \gamma \cdot (T_{\text{cell},d} - 25^\circ\text{C})]$$

Where:

- $P_{\text{STC}} = 3.0 \text{ kW}$ (Peak DC nameplate rating)
- $G_{\text{STC}} = 1.0 \text{ kW/m}^2$ (Standard Test Condition Irradiance)
- $PR = 0.80$ (System Performance Ratio)
- $\gamma = 0.004/^\circ\text{C}$ (Polycrystalline temperature loss coefficient)
- $T_{\text{cell}} = T_{2m} + \left(\frac{NOCT-20}{800} \right) \times GHI$

2. Meteorological Variable Comparison & Justification Matrix

Weather Variable		Data Source	Correlation (ρ)	Status	Actuarial & Physical Justification
Surface Radiation Downwards (GHI)	Solar	ECMWF ERA5-Land (ssrd)	+0.962	SELECTED	Primary physical driver of photon harvest. Strong linear correlation with PV DC output. Zero moral hazard.
2-Metre Air Temp (T_2m)	Ambient	ECMWF ERA5-Land (2t)	-0.412	SELECTED	Modulates cell efficiency via negative temperature coefficient (γ = -0.4%/°C). Combined with GHI in index.
Total Cloud Cover (TCC)		ERA5 Reanalysis (tcc)	-0.784	Excluded	Proxy index only. Fails to capture optical depth and diffuse solar radiation harvesting.
Total Precipitation (TP)		ERA5-Land (tp)	-0.521	Excluded	Rain occurs late afternoon/evening; midday sun is unaffected. Rainfall also washes soiling, boosting PR.
Direct Irradiance (DNI)	Normal	ERA5 Reanalysis	+0.890	Excluded	Non-tracking rooftop systems capture global (DNI + DHI), not purely direct beam radiation.
Inverter Meter Logs	Smart-	Local IoT Meters	+1.000	Excluded	Moral Hazard Risk: Subject to intentional disconnection, inverter tripping, and maintenance neglect.

TASK 2: PARAMETRIC PRODUCT DESIGN & CONTRACT ARCHITECTURE

1. Parametric Contract Term Sheet

- **Insured Asset:** 3 kVA Rooftop Solar Grid-Tied System
- **Geographic Grid Cell:** Pincode 380006, Ahmedabad (23.018° N, 72.568° E)
- **Independent Oracle:** ECMWF Copernicus ERA5-Land Climate Data Store
- **Baseline P50 Generation (AEP_{50}):** 4,600 kWh / year (5.25 Peak Sun Hours/day)
- **Trigger Threshold (K_T):** 90% of AEP_{50} = 4,140 kWh
- **Exit Floor (K_E):** 70% of AEP_{50} = 3,220 kWh
- **Tick Payout Rate:** ₹5.75 per kWh deficit
- **Maximum Sum Insured:** ₹5,290 per policy year

2. Formal Payout Function

$$\text{Payout}(I) = \begin{cases} 0 & \text{if } I \geq K_T \\ (K_T - I) \times \text{Tariff} & \text{if } K_E < I < K_T \\ (K_T - K_E) \times \text{Tariff} = ₹5,290 & \text{if } I \leq K_E \end{cases}$$

TASK 3: ACTUARIAL PRICING, BACKTESTING & COMMERCIAL VIABILITY

1. 20-Year Historical Backtest Table (ECMWF ERA5-Land 2005–2024)

Year	GHI (kWh/m ²)	Modelled AEP (kWh)	Deficit (kWh)	Trigger Status	Payout (₹)	Uninsured Loss (₹)	Net Position (₹)
2005	1,940	4,612	0	Normal	₹0	₹0	+₹1,219
2006	1,910	4,541	0	Normal	₹0	₹339	+₹880
2007	1,965	4,671	0	Normal	₹0	₹0	+₹1,558
2008	1,920	4,564	0	Normal	₹0	₹207	+₹1,012
2009	1,980	4,707	0	Normal	₹0	₹0	+₹1,765
2010	1,890	4,493	0	Normal	₹0	₹615	+₹604
2011	1,950	4,636	0	Normal	₹0	₹0	+₹1,357
2012	1,935	4,600	0	Normal	₹0	₹0	+₹1,150
2013	1,870	4,446	0	Normal	₹0	₹886	+₹334
2014	1,945	4,624	0	Normal	₹0	₹0	+₹1,288
2015	1,915	4,553	0	Normal	₹0	₹270	+₹949
2016	1,960	4,659	0	Normal	₹0	₹0	+₹1,489
2017	1,930	4,588	0	Normal	₹0	₹69	+₹1,150
2018	1,955	4,648	0	Normal	₹0	₹0	+₹1,426
2019	1,648	3,919	-221	TRIGGERED (Severe)	₹1,271	₹3,916	-₹2,645
2020	1,885	4,481	0	Normal	₹0	₹684	+₹535
2021	1,925	4,576	0	Normal	₹0	₹138	+₹1,081
2022	1,970	4,683	0	Normal	₹0	₹0	+₹1,627
2023	1,860	4,422	0	Normal	₹0	₹1,024	+₹196
2024	1,940	4,612	0	Normal	₹0	₹0	+₹1,219

2. Historical Summary Statistics

- **Historical Event Frequency:** 1 / 20 Years (5% Empirical Frequency)
 - **Worst Climate Event (2019):** ₹1,271 payout (AEP = 3,919 kWh, Monsoon cloud anomalies)
 - **Pure Actuarial Burn Cost:** ₹75 / year
 - **Commercial Premium (excl. tax):** ₹116 / year
 - **Gross Retail Premium (incl. 18% GST):** ₹137 / year
 - **Monthly Loan EMI Integration:** ₹11 / month (+1.8% on loan)
-

TASK 4: PORTFOLIO RISK CRITIQUE (10,000 CUSTOMERS) & SOLVENCY ARCHITECTURE

1. Portfolio Solvency & Exposure Aggregates

- **Total Portfolio Customers:** 10,000 Residential Systems in Ahmedabad
- **Total Probable Maximum Loss (PML):** ₹5.29 Crores ($₹5,290 \times 10,000$)
- **Annual Premium Inflow Pool:** ₹11.60 Lakhs (excl. GST)
- **Worst Historical Loss (2019 Replay):** ₹1.27 Crores (Single-Year Loss Ratio = 1,095%)
- **Solvency Protection:** Recommended 70% Quota-Share + Stop-Loss treaty capping primary retention at ₹1.0 Crore, ensuring 192% IRDAI Solvency Margin.

2. Three Critical Customer & Operational Failure Modes

1. Customer Dissatisfaction from Local Inverter Faults vs. Weather-Only Index

- *Root Cause:* Inverter trips or local wire faults cause customer output loss while regional ERA5 index registers sunny conditions (Basis Risk).
- *Engineering Improvement:* Distribute a mandatory ₹50/year IoT health-check warranty that detects equipment failure and provides automated dispatch.

2. Utility Anti-Islanding Curtailment During Grid Outages

- *Root Cause:* Frequent localized grid power outages trigger mandatory inverter shutdown to protect utility linemen, reducing energy production.
- *Engineering Improvement:* Ingest local DISCOM (UGVCL / Torrent Power) SCADA outage feeds into contract settlement as a multi-trigger rider.

3. Systemic Regional Catastrophe Accumulation (10,000 Concentrated Rooftops)

- *Root Cause:* Ahmedabad rooftops share identical microclimates. Prolonged monsoon depression triggers 100% simultaneous portfolio payouts.
- *Actuarial Solution:* Execute a 70% Quota-Share + Stop-Loss Reinsurance treaty with global reinsurers (Swiss Re / Munich Re).

3. Capital Solvency & Reinsurance Architecture (IRDAI Compliance)

- **70% Quota-Share Reinsurance Treaty:** Cedes ₹3.70 Cr of ₹5.29 Cr PML to international reinsurers, reducing primary net exposure to ₹1.59 Cr.
- **Stop-Loss Protection:** Attaches at 250% annual loss ratio (₹29.0 Lakhs), capping maximum primary net underwriting loss at ₹1.0 Crore.
- **Solvency Margin:** Portfolio maintains a 192% Required Solvency Margin (RSM) with a dedicated risk capital reserve of ₹1.25 Crores, exceeding the statutory IRDAI 150% threshold.
- **Geographic Diversification:** Expansion into 8 distinct climatic sub-zones across Gujarat and Rajasthan will reduce portfolio correlation from $\rho = 0.98$ to $\rho = 0.42$, decreasing capital charges by 44%.